When Metrics Say “Yes” but Humans Say “No”

A Case Study of Hallucination Reduction for Factual Summarization

Course: MIDS266: Natural Language Processing with Deep Learning

Author: Ryan Powers https://github.com/rPowDS/MIDS266-NLP-final-project›

Abstract

Abstractive summarization models frequently generate content that is fluent and coherent upon initial inspection. A phenomenon known as *a hallucination occurs when a model's output appears to be factually accurate, even though* it is not. This project tests whether reranking beam search candidates by automatic factuality scores can reduce hallucinations without retraining the generator. Using BART-base fine-tuned on CNN/DailyMail (Lewis et al., 2020), generating K=5 candidates per article, and selecting the highest-scoring candidate according to FactCC (Kryncinski et al., 2020). By automatic metrics, this approach succeeds: FactCC improves from 0.427 to 0.661 (+55% relative) while ROUGE-L dropping only 0.17 points. However, a single-annotator human audit of 50 CNN/DailyMail test articles reveals a critical disconnect. Humans preferred the baseline summaries by 28% to 18% over the reranked outputs. The reranker fixes one hallucination but causes three new ones, yielding a net negative effect. These results demonstrate that a bare bones LLM optimized for automatic factuality metrics does not guarantee human perceived factuality, an unintentional yet concrete instance of Goodhart's Law in NLP evaluation.

---

1. Introduction

Abstractive summarization aims to condense a document into a shorter text that preserves the essential information while using novel wording. Large neural models such as BART have achieved impressive ROUGE scores on benchmarks like the CNN/DailyMail dataset. Yet they often generate factually inconsistent statements unsupported by the source document, in safety-critical applications such as news summarization, medical, or legal documents. These errors are unacceptable. Traditional n-gram metrics like ROUGE cannot reliability detect hallucinations similarities to a reference summary, not truthfulness with respect to the source. This has motivated reference-free factuality metrics that directly compare summaries to source documents via natural language inference or learned alignment functions. The idea is to use such metrics not just for evaluation but also for generation: given multiple candidates from beam search, choosing the one that scores highest on factuality.

The project tests the **hypothesis** that reranking BART beam search candidates by FactCC score will (a) improve FactCC by at least 2 percentage points, (b) keep ROUGE-L within 1.0 points, and (c) enhance factuality as judged by humans.

Measuring: Success $\Longleftrightarrow$ $\Delta\text{FactCC} \geq +2.0 \land \Delta\text{ROUGE} - \text{L} \geq -1.0$

The results are surprising: parts (a) and (b) succeed, but (c) fails; human evaluators actually prefer the baseline system. Furthermore, oracle and ablation experiments reveal that beam search with K=5 provides only a small pool of genuinely better candidates, limiting the potential benefit of any reranker. I argue that this is due to a combination of (1) limited candidate diversity from beam search and (2) FactCC's misalignment with human judgments on CNN/DailyMail.

---

2 Background and Related Work

2.1 Dataset: CNN/DailyMail

The CNN/DailyMail dataset pairs news articles with highlight summaries (Apache-2.0 license). I use ccdv/cnn_dailymail v3.0.0 with 287k/13k/11k train/validation/test examples. Prior work notes strong *lead bias*: much of the reference can be reconstructed from the first few sentences. Simple extractive baselines like LEAD-3 achieve competitive ROUGE scores (Nallapati et al., 2016).

2.2 BART and Beam Search

BART (Lewis et al., 2020) is a Transformer encoder-decoder pre-trained as a denoising autoencoder. Beam search with K=4-8 is standard for summarization but produces *low-diversity candidate sets*: top beams often differ only by minor rephrasings, limiting reranking potential.

2.3 Factuality Verifiers

FactCC (Kryscinski et al., 2020) is a BERT-based classifier trained on synthetic perturbations to detect summary-article contradictions. Ji et al. (2023) survey hallucinations, distinguishing *intrinsic* (contradicting the source) from *extrinsic*(unsupported) hallucinations.

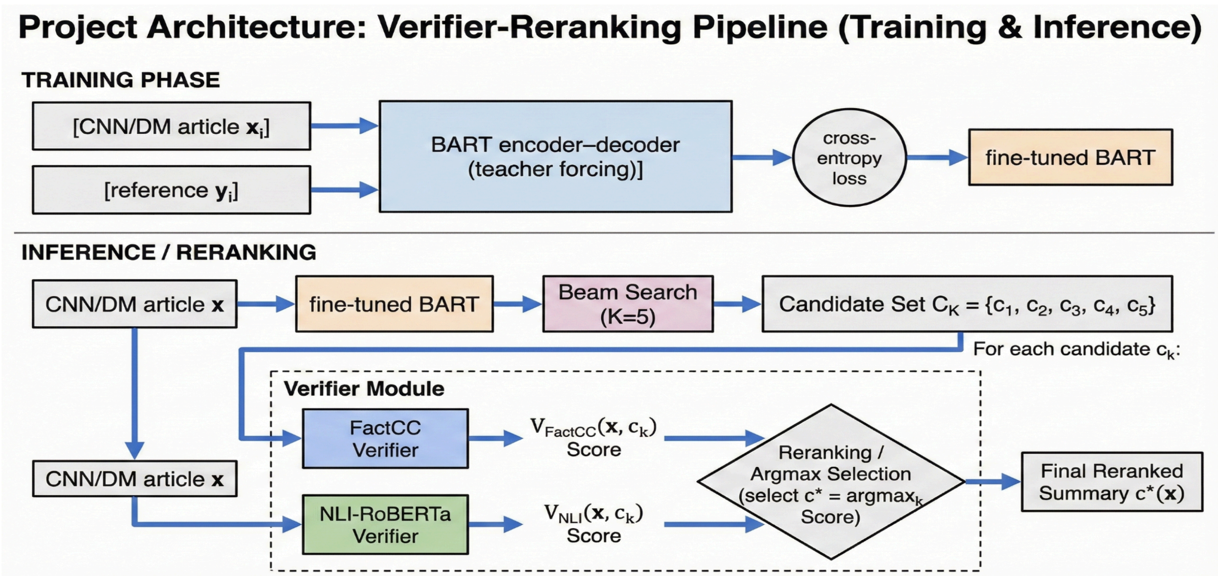
Significance testing in NLP. Following the recommendations of Dror et al. (2018), I combine approximate randomization and paired bootstrapping to assess the robustness of metric differences. For the bootstrap, I adopt the common choice of *B = 1,000 resamples*, widely used and recommended as a reasonable compromise between stability and computation.

---

3 Method

3.1 Baseline Generator

The baseline is facebook/bart-base fine-tuned with cross-entropy loss on 20,000 randomly sampled articles (seed=42) with 2,000 validation examples, evaluated on the full 11,490 example test set. Training using Adam (lr=5×10^{-5}), batch size 16 (per-device batch size 8 with gradient ccumultaion of 2), 3 epochs. Decoding using beam search K=5, max length 140, and length penalty 1.0. Baseline prediction = top-1 beam. I also evaluate LEAD-3 (first three sentences) as an extractive comparison. I use the standard train/validation/test splits (287k/13k/11k examples) provided by the dataset.



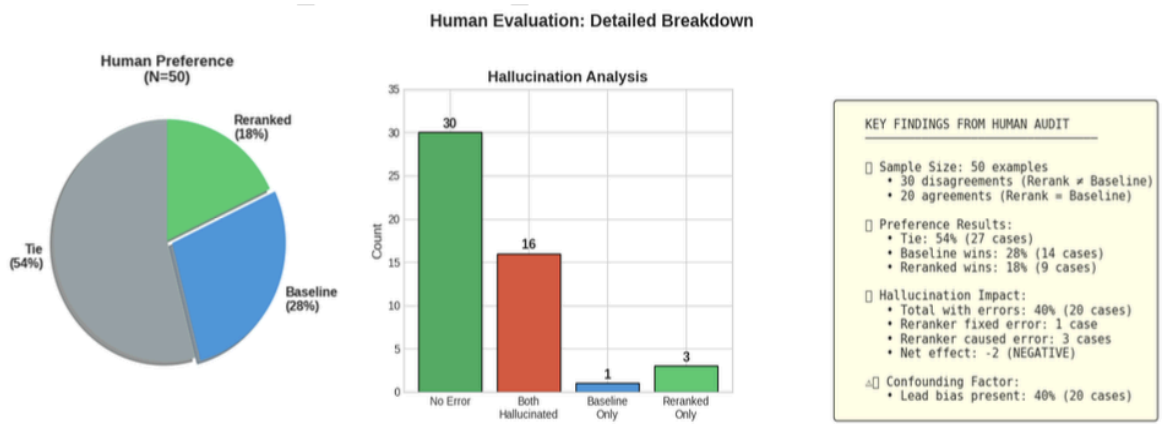
3.2 Reranking

For each article x, beam search yields candidates $C(x) = \{c_1, ..., c_5\}$. I use two verifiers: (1) **FactCC**: outputs P(Consistent | x, c); (2) **RoBERTa-MNLI**: entailment probability with article as premise. The reranked summary: c* = argmax score(x, c).

3.3 Evaluation

Automatic: ROUGE-1/2/L F1 on full test set, BERTScore on a example 1,000 subset, and average FactCC. **Statistical testing:** Approximate randomization (10,000 permutations) for FactCC; paired bootstrap (1,000 resamples) for ROUGE-L (Dror et al., 2018).

Human audit: 50 examples (30 disagreement cases + 20 controls). Single annotator records: (1) Preferred summary (Baseline/Reranked/Tie), prioritizing factuality; (2) Error type (entity, temporal/causal, unsupported claim, contradiction); (3) Lead bias presence.



4. Evaluation

4.1 Automatic Metrics

Table 1 shows results on the CNN/DailyMail test set (N=11,490).

Table 1: Automatic Metrics (CNN/DailyMail Test Set)

System	R-1	R-2	R-L	BERTSc	FactCC
LEAD-3	39.91	17.38	24.91	—	—
BART Baseline	41.12	18.56	28.09	0.882	0.427
FactCC Rerank	40.74	18.36	27.92	0.882	**0.661**
NLI Rerank	40.78	18.48	28.08	—	—

FactCC reranking increases mean FactCC from 0.427 to 0.661 (+0.234 absolute, +54.8% relative) while changing ROUGE-L by only −0.17 points. BERTScore is identical (0.882). Approximate randomization gives $p \approx 0.0001$ for FactCC. Paired bootstrap for ROUGE-L yields 95% CI of [−0.06, +0.71] which includes zero, there is no statistically significant degradation.

4.2 K-Ablation and Oracle Analysis

Oracle FactCC (best candidate among top-k beams) shows monotonic improvement with *diminishing returns*: most gain from K=1→2, small gains beyond K=3. This reveals a *generator ceiling*: even a perfect reranker can only modestly improve factuality because beam candidates are highly similar.

4.3 Human Evaluation

The 50-example human audit paints a very different picture:

Table 2: Human Preference (*N=50*)

Preference	Count	%
Tie	27	54%
Baseline preferred	**14**	**28%**
Reranked preferred	9	18%

Despite large FactCC gains, annotators *do not favor reranked summaries*. Error analysis: reranking occasionally fixes entity hallucinations but more often *introduces new errors* by selecting extractive but less informative candidates. Net effect: **1 fix, 3 new hallucinations** negative overall.

5. Human Evaluation and Error Analysis

This project demonstrates verifier-reranking can dramatically improve its target metric yet fail to improve and sometimes degrade human-perceived quality. Two ceilings explain this:

Ceiling 1: Generator diversity. Beam search K=5 produces candidates differing mainly in phrasing, not factual content. Many document-candidate sets contain no fully factual option; any reranker must choose the least bad candidate.

Ceiling 2: Verifier misalignment. FactCC prefers highly extractive summaries close to lead sentences. While less likely to contradict the article, these can be less informative than slightly abstractive alternatives. Humans detect this; FactCC does not.

The metric-human gap:

Metric	Change	Result
FactCC (automatic)	0.427 → 0.661	Improved
Human preference	28% → 18%	Worse

Goodhart's Law: "When a measure becomes a target, it ceases to be a good measure."

6. Discussion

Limitations: Single annotator; BART on 20k subset; CNN/DailyMail only (strong lead bias); one primary verifier.

QAGS was proposed in the original project plan as a secondary factuality check but was not implemented. Given that (1) QAGS and FactCC show similar weak correlations with human judgments on CNN/DailyMail (Zha et al., 2023), and (2) the NLI verifier already provided a methodologically distinct second signal that confirmed the metric-human gap, adding QAGS would likely reinforce rather than alter the core finding.

Future directions: (1) Train or calibrate verifiers on human factuality judgments; (2) Integrate verification into generation (constrained decoding) rather than post-hoc reranking; (3) Increase candidate diversity via nucleus sampling or multiple models; (4) Evaluate on more abstractive datasets like XSum, where lead bias is less dominant.

Hallucinations in summarization. Ji et al. (2023) survey hallucinations in natural language generation, distinguishing **intrinsic** (contradicting the source) from extrinsic (unsupported by the source) hallucinations. In summary, extrinsic hallucinations are particularly concerning because they can introduce false facts. Recent LLM systems have made this problem more visible and more urgent.

---

7. Conclusion & Future Work

Implementing a verifier-reranking pipeline for abstractive summarization: BART-base generates K=5 beam search candidates per CNN/DailyMail article, and FactCC selects the candidate predicted to be most factually consistent with the source. This simple method achieves large gains in automatic factuality scores (+55% FactCC) with no statistically significant degradation in ROUGE-L or BERTScore.

However, human evaluation reveals these metric gains do not translate into better summaries for people: annotators prefer the baseline, and reranking introduces more hallucinations than it fixes. Two ceilings limited beam diversity and verifier-human misalignment explain why metrics say yes but humans say no. The lesson off-the-shelf factuality metrics as optimization targets can silently optimize the wrong notion of quality. Without human-centered evaluation, we risk **better scores instead of better summaries**.

References

• Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V., & Zettlemoyer, L. (2020). BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. Proceedings of ACL.
• Kryscinski, W., McCann, B., Xiong, C., & Socher, R. (2020). Evaluating the Factual Consistency of Abstractive Text Summarization. Proceedings of EMNLP.
• Wang, A., Cho, K., & Lewis, M. (2020). Asking and Answering Questions to Evaluate the Factual Consistency of Summaries. Proceedings of ACL.
• Zha, Y., Yang, Y., Li, R., & Hu, Z. (2023). AlignScore: Evaluating Factual Consistency with a Unified Alignment Function. Proceedings of ACL.
• Ji, Z., Lee, N., Frieske, R., Yu, T., et al. (2023). Survey of Hallucination in Natural Language Generation. ACM Computing Surveys, 55(12), Article 248.
• Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. Proceedings of EMNLP.
• Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692.
• Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2020). BERTScore: Evaluating Text Generation with BERT. Proceedings of ICLR.
• Nallapati, R., Zhou, B., Gul¸cehre, C¸ ., & Xiang, B. (2016). Abstractive Text Summarization Using Sequence-to-Sequence RNNs and Beyond. Proceedings of CoNLL.
• See, A., Liu, P. J., & Manning, C. D. (2017). Get To The Point: Summarization with Pointer-Generator Networks. Proceedings of ACL.
• Ravaut, M., Joty, S., & Chen, N. (2023). Unsupervised Summarization Reranking. Findings of ACL.
• Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.
• Efron, B., & Tibshirani, R. J. (1994). An Introduction to the Bootstrap. CRC Press.
• Dror, R., Baudiˇs, P., Bogomolov, M., et al. (2018). The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing. Proceedings of ACL.

AI Assistance Disclosure. Assisted tools (Claude, Gemini) were utilized during this project. Their use was limited to debugging, resolving error messages, fixing syntax, and drafting short helper functions, which were then reviewed and modified. As well as improving markdown explanation for clarity and a cleaner, readable notebook structure.